

NPL Yield Optimization

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Target Stakeholders: Recovery Operations & Strategy (1st & 2nd Line of Defense)



Executive Summary

This interactive dashboard is a decision-support tool designed to optimize the Net Present Value (NPV) of Non-Performing Loan (NPL) portfolios. By moving beyond binary "Sell vs. Hold" choices, this model demonstrates a **Hybrid Recovery Strategy**: isolating the high-FICO "tail" for internal/legal recovery while utilizing bulk asset sales for the lower-propensity "bedrock."

Key Business Impact:

- **Yield Enhancement:** Demonstrates how the 3-to-7 year "legal tail" can outperform 8% bulk asset sales by 2.5x on an NPV basis.
- **Operational Bandwidth:** Explicitly incorporates "Cost-to-Collect" and team capacity constraints to ensure the 1st Line (Operations) is not over-leveraged.
- **Strategic Alignment:** Directly supports Citi's 2026 Transformation goals of modernizing infrastructure and enhancing data-driven performance.



Interactive Features & Executive UX

The application is built using **Python and Streamlit**, featuring a precision-grade UI designed for senior leadership review.

Tab 1: Strategic Simulator (NPV Crossover)

- **Goal:** Identify the exact point where the time-value of legal recovery exceeds immediate cash-in-hand.
- **Precision Inputs:** Includes granular `number_input` fields for exact Discount Rates and Asset Sale Percentages (e.g., 7.25%), moving beyond the friction of traditional sliders.

- **NPV Modeling:** Uses dynamic curves to visualize the impact of the bank's cost of capital over a 7-year horizon.

Tab 2: Portfolio Lab (Segmentation)

- **Goal:** Visualize the distribution of the portfolio and define the "FICO Floor."
- **Audit-Ready Data:** The "Priority Litigation Targets" table utilizes strict two-decimal currency formatting with comma separators (e.g., \$24,200.00) to mirror internal bank ledger standards.
- **Behavioral Modeling:** Incorporates a "Propensity to Pay" (P2P) score to filter high-yield targets within the chosen FICO segment.

Technical Implementation

This project demonstrates proficiency in financial engineering, data visualization, and full-stack Python deployment.

- **Languages:** Python 3.x
- **Libraries:** Pandas (Data manipulation), NumPy (Financial modeling), Plotly (Dynamic graphing), Streamlit (Application framework).
- **Structure:**
 - `app.py` : Main dashboard application logic.
 - `requirements.txt` : Environment configuration for one-click deployment.
 - `data/` : Synthetic portfolio generation engine simulating real-world NPL behaviors.

Installation & Usage

To run this dashboard locally, ensure you have Python installed and follow these steps:

1. **Clone the repository:** `git clone https://github.com/your-username/recovery-legal-strategy-portfolio.git`
2. **Install dependencies:** `pip install -r requirements.txt`
3. **Launch the application:** `streamlit run app.py`

Repository Strategy

This repository serves as a "narrative of authority," demonstrating that strategic ideas in the 2nd line of defense must be grounded in the operational reality of the 1st line.

Screenshots

Global Strategy Parameters

Type exact values below for precision modeling.

NPV Sale Price (%)
8.00 - +

Legal Recovery (%)
65.00 - +

FICO Segmentation Floor
680 - +

Duration (Years)
7 - +

Discount Rate (%)
8.00 - +

Strategic Simulator Portfolio Lab

Yield Comparison: Immediate Sale vs. Discounted Recovery

Years Post-Default	Immediate Sale (\$M)	Legal NPV (\$M)
0	14.0	13.0
1	14.0	13.5
2	14.0	14.0
3	14.0	14.5
4	14.0	15.0
5	14.0	15.5
6	14.0	16.0
7	14.0	16.5

Priority Litigation Targets

Borrower Reference ID	FICO Score	Current Balance	Propensity to Pay
ACCT-0371		684	\$23,299.58
ACCT-0542		687	\$23,238.05
ACCT-0909		801	\$22,992.88
ACCT-0064		706	\$22,852.90
ACCT-0913		760	\$22,813.87
ACCT-0955		728	\$22,785.86
ACCT-0787		682	\$22,775.90
ACCT-0071		757	\$22,771.30

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Strategic Simulator Portfolio Lab

Portfolio Distribution & Segmentation

FICO Score	SOLD Count	LITIGATE Count
450	2	0
500	5	0
550	10	0
600	20	0
650	35	0
700	50	10
750	60	20
800	55	15
850	45	10
900	35	5
950	25	2

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